

RS003 Week 7 — English Worksheet

Topic: Move the Gun Left and Right · **Course:** Pygame Turret · **Time:** about 45 minutes

This week your turret comes to life. The barrel **aims** left and right while you hold the arrow keys. You learn that the X axis goes across the screen — **left makes X smaller, right makes X bigger** — and that the base stays still while only the gun moves.

💬 Words to know: **X axis**, **key press**, **arrow key**, **increase / decrease**, **speed**

1 · Predict 🧠

Read each piece of code. Before you run it, write what you think will happen.

```
gun_x = CENTER_X
SPEED = 5
gun_x -= SPEED
```

Does `gun_x -= 5` move the gun left or right? Why?


```
keys = pygame.key.get_pressed()
if keys[pygame.K_LEFT]:
    gun_x -= SPEED
if keys[pygame.K_RIGHT]:
    gun_x += SPEED
```

What happens while you hold the LEFT key? What about holding RIGHT?


```
pygame.draw.rect(screen, DARK_GREEN, (CENTER_X - 30, HEIGHT - 80, 60, 40))
pygame.draw.line(screen, GRAY, (CENTER_X, HEIGHT - 80), (gun_x, HEIGHT - 200), 10)
```

The base uses `CENTER_X` but the barrel tip uses `gun_x`. When you press a key, which part moves and which part stays still?

2 · Run It 🏃

🎯 Type the example, run it, and aim the barrel

```
import pygame
pygame.init()

WIDTH, HEIGHT = 800, 600
CENTER_X = WIDTH // 2
screen = pygame.display.set_mode((WIDTH, HEIGHT))

SKY = (135, 206, 235)
DARK_GREEN = (0, 100, 0)
GRAY = (80, 80, 80)
BLACK = (0, 0, 0)

gun_tip_x = CENTER_X
SPEED = 5

clock = pygame.time.Clock()
running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    keys = pygame.key.get_pressed()
    if keys[pygame.K_LEFT]:
        gun_tip_x -= SPEED
    if keys[pygame.K_RIGHT]:
        gun_tip_x += SPEED

    screen.fill(SKY)
    pygame.draw.rect(screen, DARK_GREEN, (CENTER_X - 30, HEIGHT - 80, 60, 40))
    pygame.draw.circle(screen, BLACK, (CENTER_X, HEIGHT - 80), 10)
    pygame.draw.line(screen, GRAY, (CENTER_X, HEIGHT - 80), (gun_tip_x, HEIGHT - 200), 10)
    pygame.display.flip()
    clock.tick(60)

pygame.quit()
```

Hold LEFT, then RIGHT. Describe how the barrel tip moves and what stays still:

3 · Spot the Bug 🐛

Each block below was meant to do something but is broken. Fix it, then explain why the original was wrong.

Bug A — Pressing RIGHT should move the barrel right, but it moves **left** instead.

```
if keys[pygame.K_RIGHT]:  
    gun_tip_x -= SPEED
```

Hint: right means X gets bigger.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The barrel should only move **while a key is held**, but it drifts on its own without any key press.

```
gun_tip_x += SPEED  
if keys[pygame.K_LEFT]:  
    gun_tip_x -= SPEED
```

Hint: the first line runs every loop no matter what.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The keys do nothing at all. The barrel never moves.

```
for event in pygame.event.get():
    if event.type == pygame.QUIT:
        running = False
    if keys[pygame.K_LEFT]:
        gun_tip_x -= SPEED
```

Hint: `keys` was never asked for. You need `pygame.key.get_pressed()` outside the event loop.

Write the fixed code (show where `keys` is set and used):

Why was it wrong? Why does your fix work?

4 · Modify It

Change the aiming speed

Try `SPEED = 2` , then `SPEED = 10` . Run each one.

Write how the aiming feels at slow speed versus fast speed:

Keep the base still, move only the gun

Make sure your base rectangle uses `CENTER_X` and only the barrel tip uses `gun_tip_x` .

Write down which line uses `CENTER_X` and which line uses `gun_tip_x` :

Hint: the base is bolted to the ground. Only the moving end of the barrel should follow `gun_tip_x` .

5 · Make It 📷

Build an aiming turret

Build a program where:

1. the barrel aims left and right with the arrow keys,
2. the base stays still,
3. you can choose the aiming speed with a `SPEED` constant.

Send a **video** (so the movement is visible), then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I made a variable to hold the barrel position called ...

Pressing LEFT changes it by ... and RIGHT by ...

The base stays still because it uses ...

I set the speed to ... because ...

One tricky moment was when ...

If I had more time, I would ...

6 · Record Your Walkthrough

Take a video while you aim the turret with the arrow keys. Talk through it like you are teaching someone. Try to use these words: **X axis**, **key press**, **arrow key**, **increase**, **speed**.

1. Show the barrel aiming left and right as you press keys.
2. Read your two `if keys[...]` lines out loud and say which way each moves.
3. Point at the line that keeps the base still.
4. Change the SPEED live and show the difference.

Write what you will say in your video. Plan it here before you record.

Submit 

Send this worksheet + your aiming video to teacher on KakaoTalk.